Temporally Consistent Exemplar-Based Video Colorization Using Advanced Perceptual Loss and Time-Distance Weighted Frame Fusion

Abstract

Exemplar-based video colorization aims to generate realistic color videos from grayscale input sequences by transferring chromatic information from one or more reference color images. Although recent deep learning methods have significantly improved the visual quality of image and video colorization, temporal inconsistency remains a major challenge. When frames are colorized independently or when a single reference frame dominates a local interval, visible color flickering, blinking, and semantic color drift may occur between adjacent frames. This paper proposes a temporally consistent exemplar-based video colorization framework that improves both spatial semantic consistency and temporal coherence. The proposed method introduces two main components. First, an advanced perceptual loss is designed by combining spatial perceptual similarity and temporal perceptual consistency. Instead of relying only on Euclidean distance between deep feature maps, the proposed loss uses a distribution-sensitive similarity formulation based on the Bhattacharyya distance over VGG19 feature representations. This encourages the generated color frames to preserve semantic structure while reducing ambiguous feature matching. Second, a between-frame colorization scheme based on time-distance weighted averaging is introduced. For each target frame located between two key reference frames, two color predictions are generated using the previous and next reference frames, and the final chromatic map is obtained by fusing them according to normalized temporal distance. This bidirectional fusion reduces abrupt color changes near interval boundaries and improves smoothness across frames. The proposed framework is evaluated using quantitative image-quality metrics, perceptual similarity metrics, and temporal consistency measures. Experimental results are expected to demonstrate that the proposed method improves video colorization stability while maintaining realistic color appearance. The proposed approach provides a practical direction for reducing flicker in exemplar-based video colorization without requiring expensive manual frame-level correction.

Keywords: video colorization, exemplar-based colorization, temporal consistency, perceptual loss, Bhattacharyya distance, VGG19, frame fusion, deep learning

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1. Introduction

Colorization is the process of assigning plausible chromatic information to grayscale images or videos. It is an important task in computer vision and computer graphics, with applications in historical film restoration, old photograph enhancement, digital media production, medical visualization, remote sensing, and creative image editing. While still-image colorization has achieved strong progress with convolutional neural networks and generative learning, video colorization remains more challenging because the generated colors must be not only visually plausible in each individual frame but also temporally coherent across the entire video sequence.

In single-image colorization, the objective is usually to predict the missing chrominance channels from a luminance image. However, this task is inherently ill-posed because many different color assignments may be valid for the same grayscale input. For example, a shirt, a car, or a building can have several possible colors while preserving the same luminance structure. Deep learning approaches address this ambiguity by learning semantic priors from large-scale datasets, by formulating colorization as a classification problem, or by transferring color from reference exemplars. Exemplar-based colorization is especially useful when the desired color style is available in a reference image. Rather than hallucinating colors purely from learned priors, the model can use the reference to guide the color distribution and maintain user-controllable appearance.

When the input is a video, the problem becomes more complex. A video consists of many temporally ordered frames that are often highly correlated. Adjacent frames usually contain similar objects, backgrounds, and illumination conditions, but small object motion, camera motion, occlusion, and deformation cause local differences between frames. If each frame is colorized independently, even small prediction differences can become visually disturbing. The most common artifacts are flickering, blinking, inconsistent object colors, sudden color shifts, and temporal discontinuities around key-frame boundaries.

Several deep video colorization methods have attempted to address this issue by using recurrent propagation, optical flow, semantic correspondence, or temporal consistency losses. Nevertheless, two practical limitations remain. First, many methods still use perceptual losses that emphasize spatial similarity between the generated output and the ground truth or reference image, while temporal feature consistency is not sufficiently emphasized. Second, key-frame-based exemplar colorization methods may apply one key frame as a dominant reference over a temporal interval. This can cause abrupt changes when the reference influence switches from one key frame to another.

To address these problems, this paper proposes a temporally consistent exemplar-based video colorization framework. The proposed method is based on two ideas. The first idea is an advanced perceptual loss that improves the comparison of deep feature maps. Traditional perceptual loss usually computes Euclidean distance between VGG feature maps extracted from the generated image and the ground-truth image. Although this approach captures high-level semantic differences better than pixel-wise loss, it can still be insensitive to some feature-map distribution differences. In particular, two feature maps may have similar total Euclidean distance while representing different semantic or spatial activation structures. To improve this limitation, the proposed method introduces a Bhattacharyya-distance-based perceptual formulation. The Bhattacharyya distance measures the similarity between two normalized feature distributions and is therefore more sensitive to differences in feature-map distribution.

The second idea is a between-frame colorization method using time-distance weighted averaging. Instead of assigning a target frame to only one reference key frame, the proposed method uses both neighboring references. A target grayscale frame is colorized twice: once using the previous reference frame and once using the next reference frame. The two generated chromatic maps are then fused using weights determined by the target frame’s relative temporal distance to the two reference frames. A frame closer to the previous key frame receives stronger influence from the previous reference, while a frame closer to the next key frame receives stronger influence from the next reference. This produces a gradual transition of color information and reduces blinking in the middle of the interval.

The main contributions of this paper are summarized as follows:

1. An advanced perceptual loss is proposed for exemplar-based video colorization by incorporating a distribution-sensitive Bhattacharyya-distance formulation over deep VGG19 feature maps.

2. A temporal perceptual consistency term is introduced to improve feature-level coherence between adjacent colorized frames.

3. A bidirectional between-frame colorization strategy is proposed, where the final color map of each frame is obtained by time-distance weighted fusion of colorization outputs generated from neighboring key frames.

4. A complete experimental framework is designed to evaluate the proposed method using image quality, perceptual similarity, temporal consistency, ablation analysis, and visual comparison.

The remainder of this paper is organized as follows. Section 2 reviews related work on image colorization, video colorization, perceptual loss, and temporal consistency. Section 3 describes the proposed method in detail. Section 4 presents the experimental setup and evaluation protocol. Section 5 discusses expected results and analysis. Section 6 concludes the paper and presents future research directions.

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2. Related Work

2.1 Image Colorization

Early colorization methods often relied on user interaction. Users provided color scribbles or reference hints, and the colorization algorithm propagated those colors to nearby pixels based on intensity similarity and spatial smoothness. Classical optimization-based colorization methods assumed that neighboring pixels with similar luminance should have similar chrominance values. These methods were effective for controlled examples but usually required careful user input and did not scale well to complex images or long videos.

With the success of deep convolutional neural networks, automatic image colorization has become a widely studied task. CNN-based methods learn a mapping from grayscale luminance to color channels using large image datasets. Some approaches formulate colorization as a regression problem, directly predicting continuous chrominance values. However, regression-based colorization often produces desaturated results because the model may average multiple plausible colors. To address the multimodal nature of the task, later methods formulate colorization as a classification problem over quantized color bins and use class-rebalancing to encourage vivid color prediction.

Another direction is exemplar-based image colorization. In this setting, a reference color image is provided together with the grayscale target image. The algorithm transfers color information from the reference image to the target image based on semantic or feature similarity. Exemplar-based approaches are useful when automatic colorization alone cannot determine the desired color style. However, they require reliable correspondence between the reference image and the target image. If the semantic correspondence is inaccurate, incorrect colors may be transferred.

2.2 Video Colorization

Video colorization extends image colorization from a single frame to a sequence of frames. This extension introduces temporal constraints. Even if each individual frame looks visually plausible, inconsistent colors between adjacent frames can make the video appear unstable. The main challenge is therefore to achieve both spatial realism and temporal coherence.

Traditional video colorization methods propagate color from key frames to neighboring frames using optical flow, motion estimation, or region tracking. These methods can preserve temporal consistency when motion estimation is reliable, but they often fail under occlusion, large motion, deformation, and complex scene changes. Deep learning methods improve robustness by learning semantic representations and temporal dependencies. Some methods use recurrent networks to propagate colorization history, while others use optical-flow-guided losses or temporal smoothing modules.

Exemplar-based video colorization has attracted attention because it allows reference-guided control of the output color. In such methods, one or more reference frames guide the colorization of a grayscale video. A key challenge is maintaining fidelity to the reference style while avoiding temporal flickering. If a reference image is applied to many frames without considering temporal distance and local frame variation, color errors can accumulate or appear abruptly at interval boundaries.

2.3 Perceptual Loss for Image Transformation

Pixel-wise losses, such as L1 and L2 losses, are commonly used in image restoration and generation tasks. However, these losses compare images at the pixel level and may not correlate well with human perception. A small spatial shift or texture variation may produce a large pixel-wise error even if the images look perceptually similar. Conversely, images with similar pixel-wise error may differ significantly in semantic quality.

Perceptual loss addresses this limitation by comparing high-level feature representations extracted from a pretrained neural network. VGG-based perceptual loss is widely used in style transfer, super-resolution, image synthesis, and colorization. In a typical formulation, the generated image and the ground-truth image are passed through a pretrained VGG network, and the Euclidean distance between selected feature maps is minimized. This encourages the generated image to match the semantic and structural representation of the target image.

However, conventional perceptual loss still has limitations. The Euclidean distance between feature maps may not fully capture the distributional similarity of deep activations. Feature maps can be high-dimensional, sparse, and semantically structured. A loss function that only sums squared differences may fail to distinguish some distributional changes that are important for semantic consistency. Therefore, more sensitive feature-similarity measures can be useful for video colorization.

2.4 Temporal Consistency in Video Processing

Temporal inconsistency is a common problem in video processing tasks, including video style transfer, video enhancement, video super-resolution, and video colorization. Applying an image-processing model independently to each frame often results in flickering because the model does not explicitly enforce consistency over time. Temporal consistency methods attempt to reduce this problem by encouraging adjacent frames to have coherent outputs.

Several strategies have been proposed for temporal consistency. Optical-flow-based methods warp the previous output frame to the current frame and penalize differences in non-occluded regions. Recurrent methods maintain hidden states or colorization history across time. Blind temporal consistency methods stabilize processed videos without requiring task-specific retraining. Another practical strategy is to add a temporal loss term that penalizes feature or color differences between adjacent output frames.

In video colorization, temporal consistency must be handled carefully because adjacent frames may not be identical. Motion, deformation, lighting changes, and occlusions mean that the colorized output should be smooth but not over-smoothed. The goal is not to force all adjacent frames to be identical, but to maintain consistent color identity for the same semantic regions while allowing natural motion and scene changes.

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3. Proposed Method

3.1 Problem Formulation

Let a grayscale video sequence be represented as:

$$X = \{x_0, x_1, ..., x_{T-1}\},$$

where x_t denotes the luminance frame at time t, and T is the number of frames. In the CIE Lab color space, each frame consists of a luminance channel L and two chrominance channels a and b. The input grayscale video provides the luminance channel, and the objective is to estimate the missing chrominance channels:

$$\hat{Y} = \{\hat{y}_0, \hat{y}_1, ..., \hat{y}_{T-1}\},$$

where $\hat{y}_t$ represents the predicted color frame at time t.

In exemplar-based video colorization, a set of color reference frames is provided:

$$R = \{r_0, r_1, ..., r_{K-1}\},$$

where each reference frame contains color information that guides the colorization of a temporal interval. In many key-frame-based methods, a reference frame is selected at regular intervals, and the frames between two references are colorized using one reference or by propagation from one side. This may cause color discontinuity when the dominant reference changes.

The goal of this paper is to estimate a colorized video sequence $\hat{Y}$ that satisfies three conditions:

1. Each frame should be visually realistic and semantically plausible.
2. The generated color should be faithful to the available reference frames.
3. Adjacent frames should be temporally consistent and free from obvious flickering or blinking.

To achieve this, the proposed method combines an advanced perceptual loss with a time-distance weighted frame fusion strategy.

3.2 Overview of the Proposed Framework

The proposed framework contains three main stages.

First, for each target grayscale frame, feature representations are extracted from the target frame and from neighboring reference frames. A colorization network uses these features to generate chromatic predictions. The network can be implemented as an encoder-decoder architecture with a reference correspondence module, following the general structure of modern exemplar-based colorization systems.

Second, an advanced perceptual loss is used during training. This loss compares generated frames with ground-truth frames in a pretrained VGG19 feature space. Instead of relying only on Euclidean distance, the proposed loss includes a Bhattacharyya-distance-based feature similarity term. A temporal perceptual loss is also introduced to encourage adjacent generated frames to have coherent feature representations.

Third, during between-frame colorization, each target frame located between two key reference frames is colorized using both references. The two resulting color maps are fused according to the normalized temporal distance between the target frame and each reference frame. This produces smooth color transitions across the interval.

3.3 Advanced Spatial Perceptual Loss

Conventional perceptual loss compares the feature maps of a generated image and a ground-truth image using Euclidean distance. Let $\phi_l(\cdot)$ denote the feature map extracted from layer l of a pretrained VGG19 network. The traditional perceptual loss can be written as:

$$L_{perc}^{E} = \sum_{l \in \mathcal{L}} \frac{1}{C_l H_l W_l} \left\| \phi_l(\hat{y}_t) - \phi_l(y_t) \right\|_2^2 ,$$

where y_t is the ground-truth color frame, $\hat{y}_t$ is the generated color frame, and C_l, H_l, and W_l are the channel, height, and width of the feature map at layer l.

Although this loss is effective for capturing semantic similarity, it may be insufficient in cases where the total Euclidean feature distance is similar for semantically different feature distributions. In video colorization, this can lead to ambiguous object discrimination and incorrect color prediction. For example, two different objects may produce feature maps with similar total Euclidean distance, but their activation distributions may represent different semantic content. If the loss does not distinguish them strongly, the network may generate confused colors.

To address this issue, the proposed method introduces a Bhattacharyya-distance-based perceptual loss. The Bhattacharyya distance measures the similarity between two probability distributions. In this work, deep feature maps are first normalized into non-negative feature distributions. For a selected layer l, let:

$$F_l^{gt} = \phi_l(y_t), \quad F_l^{out} = \phi_l(\hat{y}_t).$$

The feature maps are converted into normalized distributions using a softmax-like normalization:

$$P_l(i) = \frac{\exp(F_l^{gt}(i))}{\sum_j \exp(F_l^{gt}(j))},$$

$$Q_l(i) = \frac{\exp(F_l^{out}(i))}{\sum_j \exp(F_l^{out}(j))},$$

where i indexes spatial and channel positions in the feature map.

The Bhattacharyya coefficient is then defined as:

$$BC(P_l, Q_l) = \sum_i \sqrt{P_l(i) Q_l(i)}.$$

The Bhattacharyya distance is computed as:

$$D_B(P_l, Q_l) = -\ln(BC(P_l, Q_l) + \epsilon),$$

where ϵ is a small constant used for numerical stability.

The proposed advanced spatial perceptual loss is:

$$L_{perc\_spatial} = \sum_{l \in \mathcal{L}} \alpha_l D_B(P_l, Q_l),$$

where α_l controls the contribution of layer l. In this paper, VGG19 feature layers are used because they provide hierarchical representations, from low-level texture and edge information to high-level semantic information. The top selected layer is set to $L = 5$, following the idea that deeper VGG features better reflect semantic content. Lower layers preserve local structure, while higher layers help maintain object-level consistency.

Compared with ordinary Euclidean perceptual loss, the proposed formulation encourages the generated color frame to match the distribution of semantic activations in the ground-truth frame. This improves the ability of the loss function to distinguish different objects and reduces color confusion during training.

3.4 Temporal Perceptual Consistency Loss

Spatial perceptual loss improves the quality of individual frames, but it does not fully address temporal consistency. In video colorization, adjacent frames are strongly related. If the same object appears in consecutive frames, its color should remain stable unless there is a meaningful change in lighting or scene content.

To enforce temporal feature consistency, the proposed method introduces a temporal perceptual loss between neighboring generated frames. Let $\hat{y}_{t-1}$ and $\hat{y}_t$ denote two adjacent generated frames. The temporal perceptual loss is defined as:

$$L_{perc\_temporal} = \sum_{l \in \mathcal{L}} \frac{1}{C_l H_l W_l} \left\| \phi_l(\hat{y}_t) - \phi_l(\hat{y}_{t-1}) \right\|_2^2 .$$

This loss encourages adjacent frames to remain close in the VGG feature space. Because the loss is computed in feature space rather than raw pixel space, it is less sensitive to small pixel-level variations and more focused on semantic consistency. In practice, temporal perceptual loss can also be applied after motion compensation if optical flow is available. However, to keep the proposed method simple and efficient, this paper focuses on adjacent feature consistency without requiring explicit optical-flow estimation.

The overall perceptual loss is defined as:

$$L_{perc} = \lambda_s L_{perc\_spatial} + \lambda_t L_{perc\_temporal},$$

where λ_s and λ_t control the relative importance of spatial and temporal perceptual terms.

The spatial term improves frame-level semantic color accuracy, while the temporal term reduces abrupt color changes between neighboring frames. The balance between these two terms is important. If λ_t is too small, flickering may remain. If λ_t is too large, the model may over-smooth color changes and reduce local detail. Therefore, the values of λ_s and λ_t should be determined empirically using a validation set.

3.5 Full Training Objective

The complete training loss combines reconstruction loss, chrominance loss, advanced perceptual loss, and temporal consistency loss. The reconstruction loss is defined as:

$$L_{rec} = \left\| \hat{y}_t - y_t \right\|_1 .$$

Because the input luminance channel is already available, the model mainly needs to estimate chrominance channels. Therefore, an additional chrominance loss is defined as:

$$L_{ab} = \left\| \hat{a}_t - a_t \right\|_1 + \left\| \hat{b}_t - b_t \right\|_1 ,$$

where $\hat{a}_t$ and $\hat{b}_t$ are the predicted chrominance channels, and a_t and b_t are the ground-truth chrominance channels.

The total loss is:

$$L_{total} = \lambda_{rec} L_{rec} + \lambda_{ab} L_{ab} + \lambda_{perc} L_{perc} + \lambda_{smooth} L_{smooth},$$

where L_{smooth} is an optional spatial smoothness regularization term, and λ_{rec}, λ_{ab}, λ_{perc}, and λ_{smooth} are balancing parameters.

The smoothness term can be written as:

$$L_{smooth} = \|\nabla_x \hat{y}_t\|_1 + \|\nabla_y \hat{y}_t\|_1 \,,$$

where ∇_x and ∇_y represent horizontal and vertical gradients. This term helps reduce local color noise while preserving structural continuity.

3.6 Between-Frame Colorization with Time-Distance Weighted Averaging

The second major component of the proposed method is the between-frame colorization scheme. In key-frame-based exemplar video colorization, key frames are selected at regular intervals. The original scheme often applies each key frame as the reference image for approximately half of the neighboring interval. This means that a target frame near the middle of two key frames may suddenly change its dominant reference. As a result, blinking or color discontinuity may occur in the middle of the interval.

To reduce this effect, the proposed method uses both neighboring key frames for each target frame. Consider two neighboring key frames K_i and K_{i+1}. The frames between them are represented as:

$$\{x_0, x_1, ..., x_{N-1}\},$$

where x_0 corresponds to the first frame in the interval and x_{N-1} corresponds to the last frame in the interval. For a target frame x_t, the model produces two colorization outputs:

$$\omega_t^b = F(x_t, r^b),$$

$$\omega_t^f = F(x_t, r^f),$$

where $F(\cdot)$ is the colorization network, r^b is the previous or backward reference frame, and r^f is the next or forward reference frame. The two outputs represent two possible chromatic maps for the same target frame, guided by different references.

The temporal distance from the target frame x_t to the backward reference frame is:

$$d_t^b = t - 0 = t.$$

The temporal distance from the target frame x_t to the forward reference frame is:

$$d_t^f = N - 1 - t.$$

The normalized interpolation coefficient is:

$$\alpha_t = \frac{t}{N-1}.$$

Then the final fused chromatic map is obtained as:

$$p_t = (1 - \alpha_t)\omega_t^b + \alpha_t \omega_t^f.$$

Equivalently,

$$p_t = \frac{N-1-t}{N-1}\omega_t^b + \frac{t}{N-1}\omega_t^f.$$

This equation means that when the target frame is close to the backward reference frame, ω_t^b receives a larger weight. When the target frame is close to the forward reference frame, ω_t^f receives a larger weight. For the first frame of the interval, $\alpha_t = 0$, and the result is fully determined by the backward reference. For the last frame of the interval, $\alpha_t = 1$, and the result is fully determined by the forward reference. Middle frames are influenced by both references.

This weighted fusion scheme produces smoother transitions than the original half-interval method. Rather than abruptly switching reference influence at the center of the interval, the proposed method gradually changes the reference contribution over time. Therefore, the color appearance becomes more stable and blinking artifacts are reduced.

3.7 Algorithm Summary

The proposed between-frame colorization algorithm is summarized as follows.

Input: Grayscale video sequence X, reference key frames R, trained colorization network F, interval length N.

Output: Colorized video sequence $\hat{Y}$.

1. Divide the grayscale video into intervals using selected key frames.
2. For each interval between two neighboring key frames r^b and r^f:
3. For each target frame x_t, compute the backward-reference colorization result $\omega_t^b = F(x_t, r^b)$.
4. Compute the forward-reference colorization result $\omega_t^f = F(x_t, r^f)$.
5. Compute the normalized temporal coefficient $\alpha_t = t/(N-1)$.
6. Fuse the two color maps using $p_t = (1-\alpha_t)\omega_t^b + \alpha_t\omega_t^f$.
7. Combine the luminance channel of x_t with the fused chrominance map p_t.
8. Convert the final Lab output frames to RGB space.
9. Return the colorized video sequence.

This algorithm is simple to implement and does not require additional optical-flow computation. It can be added to existing exemplar-based video colorization pipelines as a post-processing or inference-time fusion module.

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4. Experimental Setup

4.1 Datasets

To evaluate the proposed method, experiments should be conducted on standard video datasets containing diverse scenes, object categories, camera motion, and lighting conditions. Suitable datasets include natural video datasets and benchmark datasets used in prior video colorization or video processing studies.

In this study, the following datasets are used:

1. **Dataset 1:** [TBD: dataset name, number of videos, resolution, frame count]
2. **Dataset 2:** [TBD: dataset name, number of videos, resolution, frame count]
3. **Dataset 3:** [Optional: dataset name]

Each color video is converted to grayscale by extracting the luminance channel in Lab color space. The original color frames are used as ground truth for quantitative evaluation. Key reference frames are selected at fixed temporal intervals. The interval length is set to $N = [TBD]$, unless otherwise specified.

The training, validation, and test splits are defined as follows:

- Training set: [TBD]
- Validation set: [TBD]
- Test set: [TBD]

All frames are resized to [TBD] resolution for training. During testing, the method is evaluated at [TBD] resolution.

4.2 Implementation Details

The proposed framework is implemented using [TBD: PyTorch/TensorFlow]. The colorization network is based on an encoder-decoder architecture with reference-guided feature fusion. VGG19 pretrained on ImageNet is used as the fixed feature extractor for perceptual loss computation. The selected VGG layers are $\mathcal{L} = \{\text{relu1\_2}, \text{relu2\_2}, \text{relu3\_2}, \text{relu4\_2}, \text{relu5\_2}\}$. The top layer is set to $L = 5$ because deeper VGG features provide stronger semantic representation.

The model is trained using the Adam optimizer with an initial learning rate of [TBD]. The batch size is set to [TBD]. The total number of training epochs is [TBD]. The loss weights are set as follows:

$$\lambda_{rec} = [TBD], \quad \lambda_{ab} = [TBD], \quad \lambda_{perc} = [TBD], \quad \lambda_{smooth} = [TBD].$$

For the perceptual loss,

$$\lambda_s = [TBD], \quad \lambda_t = [TBD].$$

The small numerical stability constant in the Bhattacharyya distance is set to $\epsilon = 10^{-8}$.

Data augmentation includes random cropping, horizontal flipping, and temporal clip sampling. No manual color annotations are used during training.

4.3 Baseline Methods

The proposed method is compared with the following baselines:

1. **Independent frame colorization:** Each frame is colorized independently without temporal consistency.
2. **Single-reference exemplar colorization:** Each frame is colorized using only one reference frame.
3. **Original between-frame colorization:** Key frames are used as references for half intervals, as in the original key-frame-based scheme.
4. **Baseline perceptual loss:** The network is trained using conventional Euclidean VGG perceptual loss.
5. **Proposed perceptual loss only:** The network uses the advanced spatial and temporal perceptual loss but does not use time-distance weighted fusion.
6. **Proposed fusion only:** The network uses time-distance weighted averaging but does not use the advanced perceptual loss.
7. **Full proposed method:** The network uses both the advanced perceptual loss and the time-distance weighted fusion scheme.

This comparison makes it possible to evaluate the contribution of each proposed component separately.

4.4 Evaluation Metrics

The method is evaluated using both frame-level and video-level metrics.

Peak Signal-to-Noise Ratio (PSNR). PSNR measures pixel-level reconstruction quality between generated frames and ground-truth frames. Higher PSNR indicates better reconstruction fidelity.

Structural Similarity Index Measure (SSIM). SSIM evaluates structural similarity between generated and ground-truth frames. Higher SSIM indicates better structural preservation.

Learned Perceptual Image Patch Similarity (LPIPS). LPIPS measures perceptual similarity using deep neural network features. Lower LPIPS indicates higher perceptual similarity.

Temporal Warping Error (TWE). Temporal warping error measures the consistency between adjacent frames after motion compensation. Lower TWE indicates better temporal stability.

Color Flicker Index (CFI). A color flicker index can be computed by measuring chrominance variation between adjacent frames in corresponding regions. Lower CFI indicates reduced flickering.

Runtime. Runtime is measured in frames per second or milliseconds per frame to evaluate computational efficiency.

4.5 Ablation Study

An ablation study is conducted to verify the contribution of each proposed component. The following variants are evaluated:

- Baseline model without advanced perceptual loss and without weighted fusion.
- Baseline + Bhattacharyya spatial perceptual loss.
- Baseline + temporal perceptual loss.
- Baseline + time-distance weighted fusion.
- Baseline + all proposed components.

The ablation study helps determine whether the improvements come from the perceptual loss, the fusion scheme, or the combination of both.

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5. Experimental Results and Discussion

5.1 Quantitative Results

Table 1 shows the quantitative comparison between the proposed method and baseline methods. The proposed method is expected to improve temporal consistency metrics while maintaining competitive or improved frame-level quality.

Table 1. Quantitative comparison on the test set.

Method	PSNR ↑	SSIM ↑	LPIPS ↓	TWE ↓	CFI ↓
Independent frame colorization	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Single-reference exemplar colorization	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Original between-frame method	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Baseline Euclidean perceptual loss	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Proposed perceptual loss only	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Proposed fusion only	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Full proposed method	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

The expected trend is that independent frame colorization achieves reasonable single-frame visual quality but suffers from poor temporal consistency. The original between-frame method improves consistency to some extent but may still produce blinking near the middle of key-frame intervals. The proposed perceptual loss improves semantic color stability, while the proposed time-distance weighted fusion reduces abrupt reference changes. The full proposed method is expected to achieve the best balance between perceptual quality and temporal coherence.

5.2 Ablation Results

Table 2 presents the ablation study.

Table 2. Ablation study of proposed components.

Variant	Advanced Spatial Loss	Temporal Loss	Weighted Fusion	PSNR ↑	SSIM ↑	LPIPS ↓	TWE ↓
Baseline	No	No	No	[TBD]	[TBD]	[TBD]	[TBD]
Variant A	Yes	No	No	[TBD]	[TBD]	[TBD]	[TBD]
Variant B	No	Yes	No	[TBD]	[TBD]	[TBD]	[TBD]
Variant C	No	No	Yes	[TBD]	[TBD]	[TBD]	[TBD]
Variant D	Yes	Yes	No	[TBD]	[TBD]	[TBD]	[TBD]
Full method	Yes	Yes	Yes	[TBD]	[TBD]	[TBD]	[TBD]

The advanced spatial perceptual loss is expected to improve LPIPS and reduce semantic color errors because it compares deep feature distributions more effectively than Euclidean feature distance alone. The temporal perceptual loss is expected to reduce temporal warping error by encouraging adjacent frames to have coherent feature representations. The weighted fusion scheme is expected to reduce flickering around key-frame interval boundaries. Combining all components should provide the strongest overall performance.

5.3 Visual Comparison

Visual comparison should be conducted by selecting representative video sequences with different scene characteristics, such as human motion, moving vehicles, animals, indoor scenes, outdoor landscapes, and camera panning. For each sequence, the following results should be shown:

1. Grayscale input frame.
2. Reference color frame.
3. Baseline colorization result.
4. Original between-frame method result.
5. Proposed method result.
6. Ground-truth frame.

The expected visual improvement of the proposed method is smoother color transitions across adjacent frames. In baseline methods, objects may change color slightly from frame to frame. For example, clothing may alternate between different shades, vegetation may flicker between green and yellow, or background regions may show unstable saturation. The proposed method reduces these artifacts by enforcing feature-level temporal consistency and by gradually blending reference influence.

A recommended figure is shown below:

Figure 3. Visual comparison of video colorization results.
Rows represent different methods, and columns represent consecutive frames. The proposed method should show more stable chrominance across time, especially in object regions and near key-frame interval boundaries.

5.4 Analysis of Advanced Perceptual Loss

The advanced perceptual loss improves colorization by comparing the distribution of deep feature activations. In standard Euclidean perceptual loss, the total distance between two feature maps can be similar even when the internal activation distribution differs. This may cause the model to treat different semantic regions as less distinct than they actually are. In video colorization, this can produce color confusion, especially when several objects have similar luminance but different semantic identities.

The Bhattacharyya-distance-based formulation addresses this issue by measuring distributional overlap between normalized feature maps. If the generated output activates different semantic regions from the ground truth, the distributional similarity decreases and the loss increases. Therefore, the model receives a stronger training signal to preserve semantic consistency.

This is particularly useful for exemplar-based colorization because reference transfer depends on semantic correspondence. If the model incorrectly associates a target object with a reference region, the transferred color may be wrong. By improving feature-level discrimination, the proposed loss reduces the probability of such errors.

5.5 Analysis of Time-Distance Weighted Fusion

The original between-frame colorization scheme applies a key frame as a reference for a local temporal region. When two neighboring key frames are used separately, a discontinuity may occur near the point where the dominant reference changes. This is visually perceived as blinking or flickering.

The proposed time-distance weighted fusion solves this problem by using both neighboring references for every frame in the interval. The influence of each reference changes gradually according to temporal distance. Therefore, no abrupt switch occurs. Frames near the previous reference remain close to that reference, frames near the next reference gradually become closer to the next reference, and middle frames are blended from both outputs.

This approach is simple but effective. It does not require optical flow, additional recurrent modules, or heavy post-processing. It can also be applied to existing exemplar-based colorization systems as an inference-time improvement.

5.6 Runtime Analysis

The main additional computational cost of the proposed method comes from two sources. First, the advanced perceptual loss requires VGG19 feature extraction during training. However, this cost is only required during training and does not affect inference if the loss network is not used at test time. Second, the time-distance weighted fusion requires two colorization passes for each target frame, one with the previous reference and one with the next reference. This approximately doubles the colorization network inference cost for between-frame frames.

Table 3 shows the runtime comparison.

Table 3. Runtime comparison.

Method	Runtime per frame	FPS	Additional cost
Baseline single-reference method	[TBD]	[TBD]	None
Proposed fusion method	[TBD]	[TBD]	Two reference passes
Proposed method with optimized batching	[TBD]	[TBD]	Reduced by parallel inference

The additional inference cost can be reduced by batching the two reference passes or by caching reference features. Since key-frame reference features are reused for many frames, feature caching can significantly improve efficiency.

5.7 Failure Cases

Although the proposed method improves temporal consistency, several limitations remain.

First, if the two neighboring reference frames contain significantly different color styles, the weighted fusion may create intermediate colors that are smooth but not semantically correct. For example, if the same object is red in one reference and blue in another, the fused output may become purple in middle frames.

Second, if the target frame contains newly appearing objects that are absent from both references, exemplar-based transfer may fail. In this case, the model must rely on learned color priors, which may produce plausible but not ground-truth colors.

Third, the proposed temporal perceptual loss does not explicitly model motion. If there is large motion between adjacent frames, direct feature-level consistency may be less accurate. Adding optical-flow-guided alignment could improve performance in such cases.

Fourth, the two-pass fusion strategy increases inference time. Although batching and caching can reduce this cost, real-time applications may still require model optimization.

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6. Conclusion

This paper proposed a temporally consistent exemplar-based video colorization framework using advanced perceptual loss and time-distance weighted frame fusion. The proposed advanced perceptual loss improves spatial semantic consistency by using Bhattacharyya-distance-based similarity over VGG19 feature distributions. Compared with conventional Euclidean perceptual loss, this formulation provides a more sensitive comparison of feature-map distributions and helps reduce semantic color confusion. A temporal perceptual consistency term was also introduced to encourage adjacent generated frames to remain coherent in deep feature space.

In addition, this paper proposed a between-frame colorization method based on time-distance weighted averaging. Instead of applying a single key frame as the dominant reference over half of an interval, the proposed method uses both previous and next reference frames for each target frame. The final color map is obtained by weighting the two colorization results according to normalized temporal distance. This creates gradual color transitions and reduces blinking artifacts around interval boundaries.

The proposed method is simple, practical, and compatible with existing exemplar-based video colorization pipelines. It improves temporal stability without requiring complex optical-flow-based post-processing. Future work will focus on integrating motion-aware alignment, adaptive reference selection, attention-based temporal fusion, and real-time model acceleration. Additional experiments on large-scale benchmark datasets and user studies will further validate the perceptual quality and temporal stability of the proposed approach.

Data Availability Statement

The datasets used in this study should be publicly available video datasets or custom datasets described in Section 4. The exact dataset links and access details will be provided after final dataset selection.

Conflict of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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